

Chapter 2 – SMPS, BIOS, Booting & Troubleshooting

SMPS (Switch Mode Power Supply) – Revision

What is SMPS?

 SMPS provides **power to all components in a desktop computer**

Laptop vs Desktop:

- Laptop Adapter → **~18.5V output**
 - Desktop SMPS → Multiple voltages
-

SMPS Voltage Output (Important)

Wire Color	Voltage	Use
 Red	+5V	USB, logic circuits
 Orange	+3.3V	RAM, chipset
 Yellow	+12V	CPU, GPU
 Purple	+5V Standby	Always ON
 Green	PS_ON	Power ON signal
 Black	Ground	Return path

SMPS Connectors

24-Pin ATX Connector

 Supplies power to **motherboard**

4-Pin CPU Connector

 Supplies power to **processor (CPU)**

🕒 How Motherboard Starts (Power Flow)

🔧 Front Panel (F-Panel) Pins

These pins connect **cabinet buttons & LEDs** to motherboard.

🔗 4 Main Pairs:

- Power Button 🕒
- Restart Button 🔁
- Power LED 💡
- Hard Disk LED 🗎

💡 Hard Disk Indicator (Important Concept)

👉 In laptops & desktops:

- HDD symbol = **Cylinder icon**
- If system hangs → HDD LED stops blinking

Example:

👉 If HDD light is stuck → System freeze or disk issue

🧠 BIOS (Basic Input Output System)

★ What is BIOS?

👉 BIOS is the **brain of the motherboard**

Function:

- Controls how motherboard works
 - Starts system
 - Checks hardware
-

Booting Process (Step-by-Step)

1. Power ON 
 2. BIOS starts 
 3. Company logo appears
 4. Loading screen / cursor
 5. BIOS checks storage device
-

Booting Device

 Device from which OS loads:

- HDD / SSD / USB
-

Important:

- OS is installed in **hard disk**
 - BIOS decides **which device to boot from**
-

BIOS Stores:

- Date & Time 
 - Boot order
 - BIOS password 
-

Why BIOS is Used in Troubleshooting?

 Some issues are:

- Hardware level 
- OS level 

 Hardware issues → solved using BIOS

Chipset (Motherboard Controller)

What is Chipset?

 Controls communication between components

Old vs New Design

Old:

- Northbridge → CPU, RAM, GPU
- Southbridge → USB, HDD

New:

 Single chipset (faster)

Chipset Controls:

- USB ports
 - SATA ports
 - PCIe slots
 - Network (LAN)
 - Audio
 - CPU, RAM, GPU communication
-

Important:

 **BIOS + Chipset work together**

I/O Chip (Super I/O Controller)

What is I/O Chip?

👉 Chip with many pins (~128 pins)

Function:

- Transfers signals (like PS_ON)
 - Controls input/output devices
-

Example Flow:

1. Power button pressed
 2. PS_ON signal generated
 3. I/O chip sends signal to BIOS
-

Audio Example (Important Logic)

👉 If audio not working:

1. System starts
 2. I/O checks Power OK signal
 3. If signal missing → No audio
 4. Indicates **motherboard issue**
-

What is a Chip?

👉 Chip = **Integrated Circuit (IC)**

Function:

- Controls hardware
 - Executes commands
-

Control Flow:

👉 Audio Chip ← Chipset ← BIOS ← I/O Chip

CMOS Battery

Voltage:

👉 ~3.3V

Function:

- Stores BIOS settings
 - Keeps system time
-

Important Concept:

👉 Even when system OFF → BIOS settings saved

Boot Order

Scenario:

👉 Two Hard Disks:

- HDD1 → OS installed
- HDD2 → Only data

👉 If BIOS selects HDD2:

✗ System will NOT boot

 **Error:**

👉 “No Bootable Device Found”

Causes:

- Wrong boot order
 - HDD failure
 - OS corrupt
 - CMOS battery dead
-

CMOS Battery Issue:

👉 If BIOS resets again & again → battery dead

USB Boot Issue

👉 If USB is first boot device:

- System searches OS in USB
 - If not found → error
-

System Slow Issue

Scenario:

- OS reinstalled
- Still slow

Possible Causes:

- BIOS not updated
 - Chipset drivers missing
 - Hardware limitation
-



Detailed Boot Process

1. Power ON
2. BIOS starts
3. BIOS checks HDD
4. Looks for:
 - Windows → **bootmgr**
 - Linux/Mac → **GRUB loader**



If file not found:



Error: **No Bootable Device**



Concept:

- CPU → Processing
- RAM → Temporary space



NO DISPLAY Problem (VERY IMPORTANT)



Scenario:



Power ON but no display



Step 1: Check Caps Lock LED



Keyboard Caps Lock LED is controlled by:
BIOS → Chipset → Keyboard

Case Analysis:

Case 1: Caps Lock LED ON/OFF Working

Meaning:

- BIOS OK
- Motherboard OK
- RAM OK

Problem:

- Display issue
 - HDMI cable
 - Monitor
-

Case 2: Caps Lock NOT Working

Meaning:

- BIOS not working OR
 - RAM faulty OR
 - Motherboard issue
-

Root Cause:

- RAM failure
 - BIOS failure
-

BIOS Testing:

Check BIOS chip first pin → should get **3.3V**

- No voltage → BIOS faulty
-

NO POWER Problem (VERY IMPORTANT)

Scenario:

 System not turning ON

Step-by-Step Troubleshooting:

1 Check Power Supply

- Plug & switch ON
-

2 Desktop Check (SMPS Test)

 Use **shorting method**:

- Connect Green + Black wire

 Result:

- Fan spins → SMPS OK
 - No spin → SMPS faulty
-

3 Laptop Check

 Use multimeter:

- Output should be ~18.5V
-

4 LED Behavior

 Charger LED OFF after plug:

- Motherboard short circuit
-

NO POWER vs NO DISPLAY

Problem Meaning

No Power SMPS / Power issue

No Display RAM / BIOS / Display issue

Interview Tip:

👉 If you explain both properly → interviewer knows:

- You understand **hardware deeply**
-

Real IT Support Mindset

- ✓ Always start with **power check**
 - ✓ Observe LED indicators
 - ✓ Think step-by-step
 - ✓ Do not guess → verify
 - ✓ Identify root cause fast
-

🔍 Final Understanding

👉 Boot Process:

Power → BIOS → Chipset → HDD → OS

👉 Problem Types:

- Power issue ⚡
 - Display issue 🖥️
 - Boot issue 💾
-

Chapter 2 (SUMMARY)

SMPS (Power System)

 SMPS ka kaam:

- Computer ke sab components ko **proper voltage dena** 
- AC → DC convert karna

Important Voltages:

-  12V → CPU, GPU
 -  5V → USB
 -  3.3V → RAM
 -  5V Standby → Always ON
-

Power Flow (VERY IMPORTANT)

 System ON ka process:

1. Power button press 
 2. PS_ON signal (Green wire)
 3. I/O chip signal send karta hai
 4. BIOS start hota hai
 5. System ON
-

BIOS (Brain of Computer)

 BIOS ka kaam:

- System start karna
- Hardware check karna
- Boot device select karna

👉 BIOS store karta hai:

- Date & Time 🕒
 - Boot order 💾
 - Password 🔒
-

🔄 Booting Process

👉 Step-by-step:

1. Power ON ⚡
2. BIOS start 🧠
3. Hardware check
4. HDD/SSD check
5. OS load

👉 Boot loader:

- Windows → bootmgr
 - Linux → GRUB
-

⚙️ Chipset

👉 Chipset controls:

- CPU, RAM, GPU communication
- USB, SATA, PCIe

👉 Modern system:

👉 Single chipset (fast performance)

I/O Chip (Super I/O)

👉 Work:

- Power signal control
 - Keyboard, fan, ports control
-

CMOS Battery

👉 Function:

- BIOS settings store karta hai
- Time maintain karta hai

👉 Problem:

- Battery dead → BIOS reset
-

Boot Order Concept

👉 BIOS decides:

- Kaunsa device se OS load hoga

👉 Wrong boot order:

❌ No Bootable Device

NO DISPLAY (Important)

✅ Case 1: Caps Lock Working

👉 Meaning:

- BIOS OK
- RAM OK

👉 Problem: Display

✖ Case 2: Caps Lock NOT Working

👉 Meaning:

- RAM issue
 - BIOS issue
 - Motherboard issue
-

🔌 NO POWER (Important)

👉 Check:

1. SMPS
2. Adapter
3. Motherboard

👉 SMPS Test:

- ✓ Green + Black short
 - ✓ Fan spin = OK
-

🧠 Final Logic

👉 Types of problems:

- Power Issue ⚡
 - Display Issue 🖥️
 - Boot Issue 💾
-

CONCLUSION

👉 Real IT Support Engineer ka mindset:

- ✓ Start with power check
- ✓ Observe LEDs
- ✓ Follow step-by-step process
- ✓ Identify root cause (guess nahi karna)

👉 Final line:

“Troubleshooting is not guessing, it is logical thinking + observation” 🔥



Q & A

◆ SMPS QUESTIONS

? Q1: What is SMPS?

👉 Answer:

SMPS (Switch Mode Power Supply) is a device that converts AC power into DC power and supplies different voltages to computer components like CPU, RAM, and motherboard.

? Q2: What are SMPS voltage outputs?

👉 Answer:

- 12V → CPU
 - 5V → USB
 - 3.3V → RAM
 - 5V Standby → Always ON
-

? Q3: What is PS_ON signal?

👉 Answer:

PS_ON is a signal sent by motherboard to SMPS to start power supply. It is controlled through **Green wire**.

? Q4: How do you test SMPS?

👉 Answer:

1. Disconnect SMPS
 2. Short Green + Black wire
 3. Check fan
 - 👉 Fan spins = SMPS working
-

◆ BIOS QUESTIONS

? Q5: What is BIOS?

👉 Answer:

BIOS (Basic Input Output System) is firmware that initializes hardware and starts the booting process.

? Q6: What does BIOS store?

👉 Answer:

- Date & Time
 - Boot order
 - Password
-

? Q7: Why BIOS is important?

👉 Answer:

Because without BIOS, system cannot start or detect hardware.

◆ BOOTING QUESTIONS

? Q8: What is booting?

👉 Answer:

Bootting is the process of starting a computer and loading the operating system from storage into RAM.

? Q9: What is boot device?

👉 Answer:

Device from which OS loads (HDD, SSD, USB).

? Q10: What is boot loader?

👉 Answer:

A program that loads OS:

- Windows → bootmgr
 - Linux → GRUB
-

◆ CMOS QUESTIONS

? Q11: What is CMOS battery?

👉 Answer:

A small battery that stores BIOS settings and system time.

? Q12: What happens if CMOS battery is dead?

👉 Answer:

- BIOS resets
 - Date/time incorrect
 - Boot issues
-

◆ CHIPSET & I/O QUESTIONS

? Q13: What is chipset?

👉 Answer:

Chipset manages communication between CPU, RAM, storage, and peripherals.

? Q14: What is I/O chip?

👉 Answer:

I/O chip controls input-output operations and power signals like PS_ON.

◆ TROUBLESHOOTING QUESTIONS (VERY IMPORTANT)

NO DISPLAY

? Q15: System ON but no display, what will you do?

👉 Answer:

Step-by-step:

1. Check monitor & cable
2. Check Caps Lock LED
3. If working → display issue

4. If not working → RAM/BIOS issue

NO POWER

? Q16: System not turning ON?

👉 Answer:

1. Check power supply
 2. Test SMPS
 3. Check motherboard
 4. Check adapter (laptop)
-

BOOT ISSUE

? Q17: No Bootable Device Found?

👉 Answer:

Possible causes:

- Wrong boot order
 - HDD failure
 - OS corrupt
 - CMOS battery issue
-

? Q18: System slow after OS install?

👉 Answer:

- Install drivers
- Update BIOS
- Check hardware limitation

? Q19: HDD LED not blinking?

👉 Answer:

👉 System hang or disk issue

? Q20: Charger LED OFF after plug?

👉 Answer:

👉 Motherboard short circuit

? Q21: Difference between No Power & No Display?

👉 Answer:

No Power ⚡ No Display 💻

System not ON System ON but no screen

SMPS issue RAM/BIOS issue

? Q22: How do you troubleshoot effectively?

👉 Answer:

- Start with basics
 - Follow steps
 - Check indicators
 - Verify results
-